



Master of Data Analytics
CPSC 620 – Agile Software Development (Section 3)
Course Syllabus
Spring 2026

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Office	4th Floor, 4342 Queen Street, Niagara Falls, ON, Canada
Office Phone	N/A
Office Hours	Wednesdays, 11:30 am to 12:30 pm, on campus / by Appointment
Class Time and Location	Mondays, 12:45 pm to 2:45 pm Room # UNF 428

COURSE DESCRIPTION

This course introduces agile software development practices to big data analytics. The agile approach involves discovering requirements and delivering solutions through the collaborative effort of self-organizing and cross-functional teams and their clients. It advocates adaptive planning, evolutionary development, early delivery, and continual improvement, and it encourages flexible responses to change. It is the predominant guiding methodology in today's data analytics workplace.

LEARNING OUTCOMES AND OBJECTIVES

Upon the successful completion of this course, students should be able to:

1. Explain a data analyst's role in an agile solutions development team.
2. Demonstrate an understanding of the agile mindset and its underlying principles, such as pragmatism, the power of choice, and adapting to context.
3. Explore the fundamentals of agile and lean, and how to use both approaches to produce business value.
4. Isolate and utilize multiple agile and lean techniques.
5. Develop and display awareness of how to put these techniques into action while ensuring effective implementation.
6. Isolate and explore the challenges in applying the agile method to big data analytics and change management.

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LEARNING RESOURCES

Type	Author(s)	Title	Publisher
Required Resources			
Recommended Resources	Paul Flewelling	The Agile Developer's Handbook : Get More Value from Your Software Development: Get the Best Out of the Agile Methodology	Packt Publishing
Customized course pack			
Online learning modules are curated from a variety of sources		GitHub Docs https://docs.github.com/en	

COURSE REQUIREMENTS AND GRADING

#	Component	Individual/Pair/Group	Week Due	Weight
1.	Assignment 1	Group	Week 3	15%
2	Quiz 1	Individual	Week 5	10%
3	Quiz 2	Individual	Week 7	10%
4	Assignment 2	Group	Week 8	15%
5	Final Project	Group	Week 10	40%
6	Self and Peer Assessment	Individual	Week 10	10%
Total				100%
Quizzes [% Total]: 20%				



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Assignments [% Total]: 70%

Self and Peer Assessments [% Total]: 10%

Please Note:

This class will use the flipped classroom format where you are required to review the materials of the module before coming to class. Class participation is strongly encouraged to help you understand the materials.

LATE SUBMISSIONS

Assignments submitted late without a valid reason or prior approval will incur the following penalties:

- Submissions that are **one day late** will receive a deduction of 15 percent.
- Assignments submitted **two to three days late** will receive a 25 percent deduction.
- Work submitted **four to five days late** will be subject to a 35 percent deduction.
- Assignments submitted **six to seven days late** will automatically receive a grade of zero.

Extensions may be granted in advance under extenuating circumstances.

COURSE GRADING AND SCALE

All final grades and GPAs are assigned and calculated according to the undergraduate and graduate grading scales. Each letter grade used at the University has a corresponding numeric value which is used to calculate grade point averages.

Definition	Standard of Evidence	Percentage	Letter Grade	Numeric Value
A – Excellent: Considerable evidence of original thinking; analysis and synthesis; extensive	Exceptional. Expertise in all learning outcomes.	90% - 100%	A+	4.33
	Outstanding. Expertise in some learning outcomes and mastery of most.	85% - 89%	A	4.00

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knowledge base; initiative; and fluency of expression	Excellent. Mastery of most learning outcomes, expertise in some.	80% - 84%	A-	3.67
B – Good: Clearly above average performance with knowledge of principles and facts generally complete	Very Good. Mastery of all learning outcomes.	76% - 79%	B+	3.33
	Good. Mastery of most learning outcomes, competent in some.	72% - 75%	B	3.00
	Competent. Competent in most learning outcomes, mastery of some.	68% - 71%	B-	2.67
C – Satisfactory: Evidence of some understanding of the subject matter and ability to develop solutions to basic problems.	Satisfactory. Competent in all learning outcomes.	60% - 67%	C	2.00
F – Fail: Knowledge of principles and facts is fragmentary.	Fail. No basic ability in most learning outcomes.	0 - 59%	F	0.00

You are strongly encouraged to review the rubrics of each assessment to identify what is specifically required for each assessment.

CASCADE

Students must access Cascade regularly to be successful in this course.

Cascade is the Learning Management System used by the University and can be accessed by logging into MyUNF through their student account. All activities including course content, readings, quizzes, assessments, discussion boards, assignments etc., will be available on Cascade. Contact information for student help with Cascade: cascade@myunfc.ca.

COURSE SCHEUDLE

MODULE 1	Introduction to Agile Software Development and the Agile Manifesto
MODULE 2	Agile Software Delivery Methods

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MODULE 3	Version Control and Collaboration with GitHub
MODULE 4	Implementing Scrum in Your Team
MODULE 5	Agile User Requirements and User Stories
MODULE 6	Team Setup with Liftoffs
MODULE 7	Agile Metrics for Success
MODULE 8	Technical Practices in Agile
MODULE 9	Feedback Loops in Software Development
MODULE 10	Product Roadmaps and Delivery
The modules will be delivered in person, weekly.	
The course schedule may be revised or amended due to unusual circumstances such as instructor illness, Public Health/university requirements, or campus closure due to inclement weather. This may include providing lectures or lab material online rather than in person. Any such change, including changes in assignment dates, will be communicated to students by e-mail and/or notices on Cascade.	

UN SUSTAINABLE DEVELOPMENT GOALS



The University of Niagara Falls Canada is a signatory of the UN Sustainable Development Accord and, as such, has aligned the United Nations Sustainable Development Goals with our curriculum portfolio.



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UNIVERSITY STATEMENTS, POLICIES, GUIDELINES and
RESOURCES FOR SUPPORT

Territorial Acknowledgement

University of Niagara Falls Canada acknowledges that the land we gather on is located on the traditional territory of the *Haudenosaunee*, *Hatiwendaronk* and *Anishinaabe* peoples. We also acknowledge the many other First Nations, Métis, and Inuit peoples who call this region home.

We commit to building relationships based on respect, reciprocity, and reconciliation as we work, learn, and gather on this land. Furthermore, we acknowledge that the Niagara Region is situated on treaty land, and we stand with all Indigenous peoples, past and present, in promoting the wise stewardship of the lands on which we live.

Use of Plagiarism Detection Software

This course requires you to submit some material in electronic form. When this requirement is in place, it will be noted in the course outline, the assignment, or both. The electronic material will be submitted to a service to which the University subscribes, called Turnitin. This is a service that checks textual material for originality. Turnitin is used increasingly in North American universities.

Academic Integrity

While completing your assignments, you have the freedom to use any tools or resources offered by the library to complete your assignments. The use of Artificial Intelligence must be approved by your instructor, and should only be used to augment your work, and must be cited in your assessments just as any other source. All citations should follow the APA format.

AI Usage

The University of Niagara Falls Canada is committed to the use of AI guided by ethical and transparent principles. While students may be allowed to use advanced automated tools (such as ChatGPT or Dall-E 3) for certain written or picture-based assignments, it is crucial to adhere to the following guidelines:

Seek prior approval from the instructor for AI use in specific assignments when the use of AI is allowed. If allowed, you will be required to attribute and cite any AI-generated content in your work, including prompts and AI outputs as part of your academic record. Include an additional reflection component in your assessments, discussing how AI tools contributed to your learning process.



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IMPORTANT: Please note that AI use is strictly prohibited in assessments and assignments not approved by the instructor. Failure to abide by this guideline may be considered an act of cheating and a violation as outlined in the relevant sections of The University of Niagara Falls Canada “**Academic Integrity & Honesty Policy 2024.**”

Attendance

Class attendance and punctuality are essential for academic success. Student attendance and active participation in all scheduled classes, grading events and activities is important for ensuring an effective learning experience for all students. UNF understands that there are circumstances such as illness, injury, family crises, or other circumstances beyond a student’s control that may prevent students from attending or participating in all classes. A student who is not officially registered in a course may not attend or participate in classes, either on campus or online.

Instructors are responsible to communicate specific attendance expectations of a course in the Course Syllabus and distribute the syllabus in the first week of classes.

Student attendance or absence in an online course is determined by the instructor according to weekly course activities and requirements outlined in the Course Outline. Student attendance in an online course is defined as active participation and submission of assignments in the course as described in the course outline. Online courses typically have weekly assessments of student participation such as discussion questions, and completion and submission of assignments to track student attendance and participation.

Absence(s) from a class or major grading event for health reasons or other extenuating circumstances must be reported to the instructor immediately and the appropriate medical or other supporting documentation from a recognized professional provided. The medical documentation will be maintained in the student file in the Registrar’s Office. Medical or other documentation for absences must be provided before the end of the term unless there are extraordinary circumstances.

A student who is unable to attend class or to participate in an examination or study requirements on a particular day due to religious beliefs must inform the instructor of any intended absences for religious observances within the first week of the term so that accommodations can be made in advance.

A student:

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- May not be permitted to attend a given class period or scheduled examination for reasons of lateness, misconduct, or failure to meet the responsibilities of the course;
- Who has been absent for acceptable reasons with appropriate documentation may receive a Deferred grade and enter into an Incomplete Course Contract with the course instructor. Incomplete Course Contracts must be approved by the Associate Dean.

Classroom Attendance and Etiquette

To be respectful of faculty and students, each student should plan to arrive prior to the class start. It is disrespectful to others to disrupt classes by arriving late. A faculty member may refuse admission to a student to a given class period or scheduled examination for reasons of lateness or misconduct. Students are expected to provide clear and acceptable reasons for any prior absence upon returning to class and are expected to notify the faculty member by e-mail at least two hours in advance when they know they will not be able to attend a given class.

Electronic Devices on Campus

Use of technology by students and instructors is permitted during class provided it is used for educational purposes. Technological devices may include laptops, tablets, cell phones, etc. Instructors are permitted to regulate the use of technology for social communication purposes. Images, videos and audio recordings of instructors or in-class activities are not permitted without prior consent of the instructor, students in class, Associate Dean or Academic Affairs.

Review of Grades

For complete policy regulations and procedures, refer to the **UNF Final Grade Reviews and Appeals Policy and Procedures**.

Most concerns regarding final grades can be successfully resolved through informal processes such as discussions with the instructor. Requests for reviews and appeals of Final Grades will be submitted in a timely manner as indicated by the Informal Grade Review procedures and Formal Grade Appeal procedures. Responses to Informal Grade Reviews and Formal Grade Appeals will be provided to students in a timely manner as provided by the procedures in this Policy.

To file a Formal Final Grade Appeal, the student must present the following information:

- The final grade the student received in the course.
- Any paper or written examinations, assignments or other documents relevant to the appeal including the documents from the two informal appeals, submissions, forms and outcomes.
- The grounds the student relies upon in making the appeal.



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- The resolution sought by the student.

A processing fee of \$50 for each final grade being appealed which shall be returned to the student if the appeal is granted.

Student Code of Conduct

For complete policy regulations and procedures, refer to the UNF Student Conduct Policy.

University of Niagara Falls Canada (UNF) is committed to developing a sense of community dedicated to a working and learning environment of the highest quality. The environment is characterized by mutual respect, consideration, social and moral development of its members, and is free from harassment, discrimination based on prejudice, and any form of physical violence.

Students are responsible for adhering to this policy as the foundation of their relationship with peers, faculty, staff, and the UNF community in general. As well as demanding adherence to the overall requirements of the laws of Ontario and Canada, students are accountable for, but not limited to:

- The exercise of self-discipline, accountability and judgment in daily interpersonal relationships as well as academic activities;
- Acceptance of personal responsibility for continued academic and professional competency and learning;
- Acceptance that one's professional abilities, personal integrity, and treatment of others as measures of mature, civil conduct reflect on UNF itself; and
- Personal accountability for any conduct which is not in keeping with reasonable university standards.

Students are responsible for their own behaviour while on campus or in the online learning environment and while participating in any university-related activities, such as meetings and workplace learning.

All students are responsible for understanding and adhering to UNF policies.

Students assume an obligation to conduct themselves at all times as responsible members of the university community, to respect personal property and the rights of others, and to support the educational mission of UNF. The University insists that students demonstrate academic and professional integrity in addition to academic excellence. The University expects all parties to resolve conduct issues using informal means before escalating to formal measures.



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Behavior which does not respect the rights and dignity of others may be considered student misconduct and is subject to the rules governing student misconduct. This policy applies to any University class, activity, function or event sponsored or organized by the University, whether on or off the University's premises or virtually.

Academic Standing and Progression

To progress to the next level of study, students must meet the following criteria:

- Achieve a minimum passing grade of 60% in each course
- Maintain a semester Grade Point Average (GPA) of at least 3.0.

The University of Niagara Falls Canada Academic Rules and Regulations

All UNF courses are governed by the academic rules and regulations set forth in The UNF Academic Calendar as approved by the UNF Academic Council and UNF Board.

The Academic Calendar is the University of Niagara Falls Canada's (UNF's) official statement of regulations, policies, programs, requirements, and course descriptions for the 2023-2024 academic year.

Associate Dean Review

Name: Hassan Baz Chamas

Signature:

A handwritten signature in black ink, appearing to read 'Hassan Baz Chamas', written over a horizontal line.

Date: March 30, 2026